

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics Extension 2

HSC ASSESSMENT TASK
JUNE 2006

General Instructions

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions

NAME : _____

QUESTION 1	QUESTION 2	QUESTION 3	TOTAL

QUESTION ONE (16 marks)

Marks

- a) Find $\int \cos^3 x \, dx$ 3
- b) Use partial fractions to find $\int \frac{24}{x^2 + 4x - 12} \, dx$ 3
- c) Find $\int \frac{24}{x^2 + 6x + 18} \, dx$ 2
- d) Evaluate $\int_0^{\frac{\pi}{6}} x \cos x \, dx$ 3
- e) Solve the equation $x^3 - 5x^2 + 11x - 15 = 0$ 2
given that $x = 1 - 2i$ is a solution.
- f) For the polynomial $P(x) = x^3 + (k + 3)x^2 + (2k + 7)x + (k + 5)$ 3
where k is real, find the possible values of k given that $x = -1$
is the only real root of $P(x) = 0$.

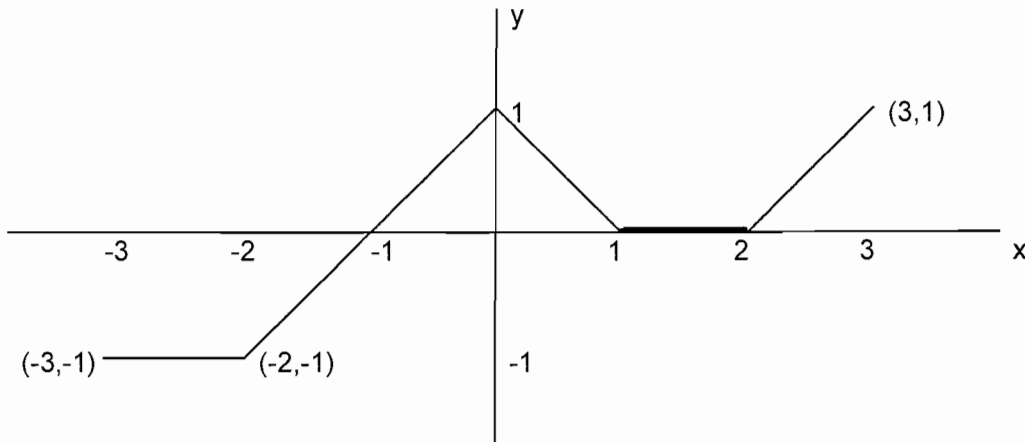
QUESTION TWO (17 marks) (Start a new page)

Marks

a) Find $\int \frac{x+1}{\sqrt{x-1}} dx$

2

b) The diagram below is a sketch of the function $y = f(x)$ defined for $-3 \leq x \leq 3$



On separate diagrams sketch

i) $y = \frac{1}{f(x)}$

2

ii) $y = \log_e f(x)$

2

iii) $y^2 = f(x)$

2

c) i) Use the identity $\cos 4\theta = 8\cos^4 \theta - 8\cos^2 \theta + 1$

3

to solve the equation $16x^4 - 16x^2 + 1 = 0$

ii) Use the above solutions to show that $\cos \frac{\pi}{12} \cos \frac{5\pi}{12} = \frac{1}{4}$

3

Justify your answer

iii) By solving the equation $16x^4 - 16x^2 + 1 = 0$

3

using a different method from above,

find the exact value of $\cos \frac{\pi}{12}$.

Justify your answer.

QUESTION THREE (17 marks) (Start a new page)

Marks

a) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$ using the substitution $t = \tan \frac{x}{2}$ 3

b) Solve $4x^3 + 4x^2 - 7x + 2 = 0$ 3
given that two of the roots are equal.

c) The equation $x^3 + px + q = 0$ (p, q are real) has roots α, β and δ .

i) Find the polynomial equation with roots $3\alpha, 3\beta$ and 3δ . 2

ii) Show that $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\delta^2}{q}$ 1

iii) Find a cubic polynomial equation with roots 2

$$\frac{1}{\alpha} + \frac{1}{\beta}, \frac{1}{\beta} + \frac{1}{\delta} \quad \text{and} \quad \frac{1}{\alpha} + \frac{1}{\delta}$$

d) i) Show that $(1-x)^{n-1} - (1-x)(1-x)^{n-1} = x(1-x)^{n-1}$ 1

ii) If $I_n = \int_0^1 \sqrt{x}(1-x)^n dx$ where n is a non negative integer

Show that $I_n = \frac{2n}{2n+3} I_{n-1}$ 3

iii) Evaluate $\int_0^1 \sqrt{x}(1-x)^3 dx$ 2

SOLUTIONS 2006

Q1. a. $\int \cos^3 x \, dx$

$$= \int \cos x \cos^2 x \, dx$$
$$= \int \cos x (1 - \sin^2 x) \, dx$$

let $u = \sin x$
 $du = \cos x \, dx$

$$= \int (1 - u^2) \, du$$
$$= u - \frac{1}{3} u^3$$
$$= \sin x - \frac{1}{3} \sin^3 x + c$$

b. $\frac{24}{x^2 + 4x - 12} = \frac{A}{x+6} + \frac{B}{x-2}$

$$\therefore 24 = A(x-2) + B(x+6)$$

when $x = -6$

$$24 = -8A$$

$$A = -3$$

when $x = 2$

$$24 = 8B$$

$$B = 3$$

$$\therefore \int \frac{24}{x^2 + 4x - 12} \, dx$$
$$= \int \frac{3}{x-2} - \frac{3}{x+6} \, dx$$
$$= 3 \ln(x-2) - 3 \ln(x+6) + c$$
$$= 3 \ln\left(\frac{x-2}{x+6}\right) + c$$

$$\begin{aligned}
 c. \quad & \int \frac{24}{x^2 + 6x + 18} dx \\
 &= \int \frac{24}{(x+3)^2 + 9} dx \\
 &= \frac{24}{3} \tan^{-1} \frac{x+3}{3} + c \\
 &= 8 \tan^{-1} \left(\frac{x+3}{3} \right) + c
 \end{aligned}$$

$$\begin{aligned}
 d. \quad & \int_0^{\frac{\pi}{6}} x \cos x dx \\
 &= x \sin x \Big|_0^{\frac{\pi}{6}} - \int_0^{\frac{\pi}{6}} \sin x dx \\
 &= \frac{\pi}{12} + \left[\cos x \right]_0^{\frac{\pi}{6}} \\
 &= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1
 \end{aligned}$$

$$\begin{aligned}
 e. \quad & \text{roots } 1-2i, 1+2i, \alpha \\
 & \text{sum of roots} = 5 \\
 \therefore & 1-2i + 1+2i + \alpha = 5 \\
 & \alpha = 3 \\
 \therefore & \text{Solutions are } 1-2i, 1+2i, 3
 \end{aligned}$$

$$\begin{array}{r}
 f. \quad \begin{array}{r}
 x^2 + (k+2)x + (k+5) \\
 x+1 \overline{) \begin{array}{r}
 x^3 + (k+3)x^2 + (2k+7)x + (k+5) \\
 \underline{x^3 + x^2} \\
 (k+2)x^2 + (2k+7)x + (k+5) \\
 \underline{(k+2)x^2 + (k+2)x} \\
 (k+5)x + (k+5) \\
 \underline{(k+5)x + (k+5)} \\
 0
 \end{array}
 \end{array}
 \end{array}$$

$$\therefore P(x) = (x+1)(x^2 + (k+2)x + (k+5))$$

quadratic has no real roots

$$\therefore \Delta < 0$$

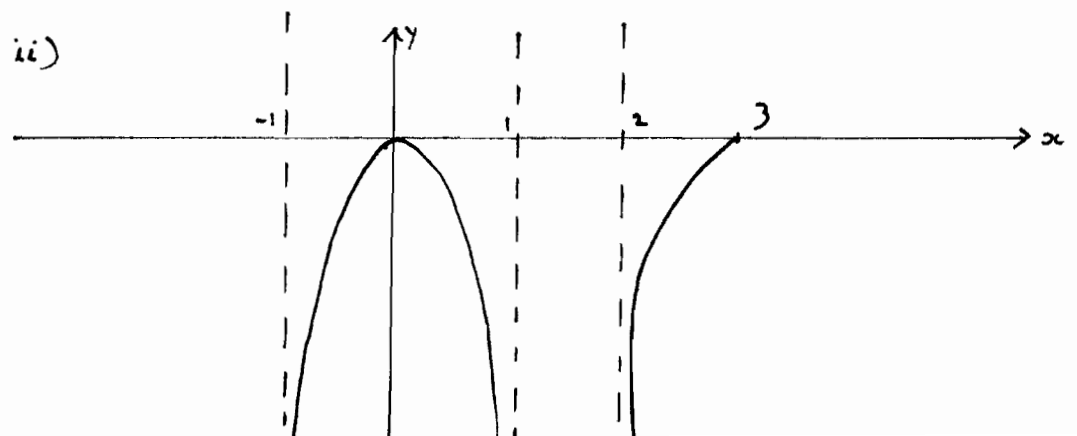
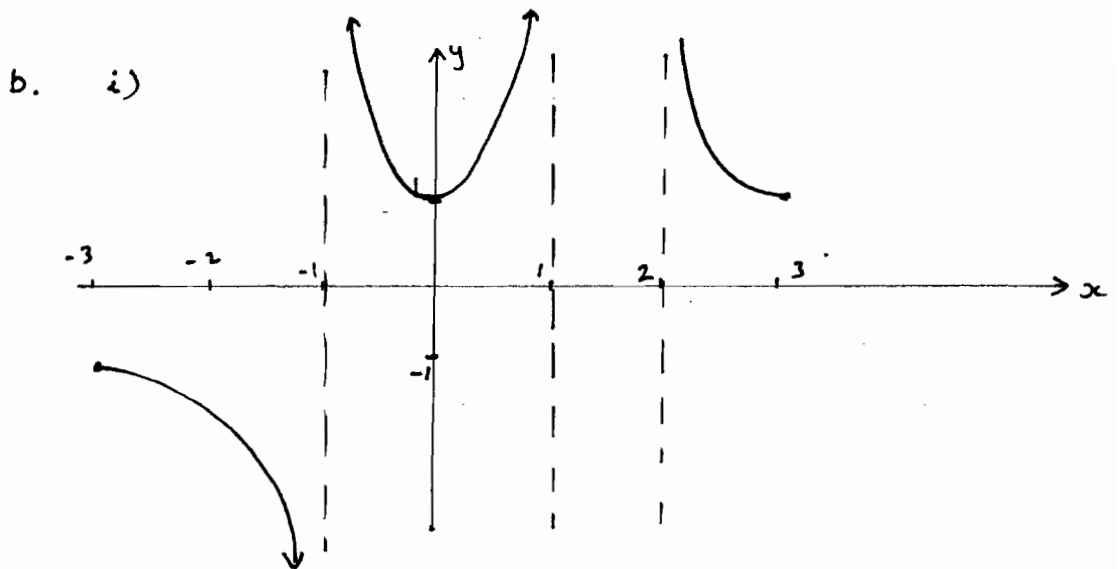
$$\therefore (k+2)^2 - 4(1)(k+5) < 0$$

$$k^2 - 16 < 0$$

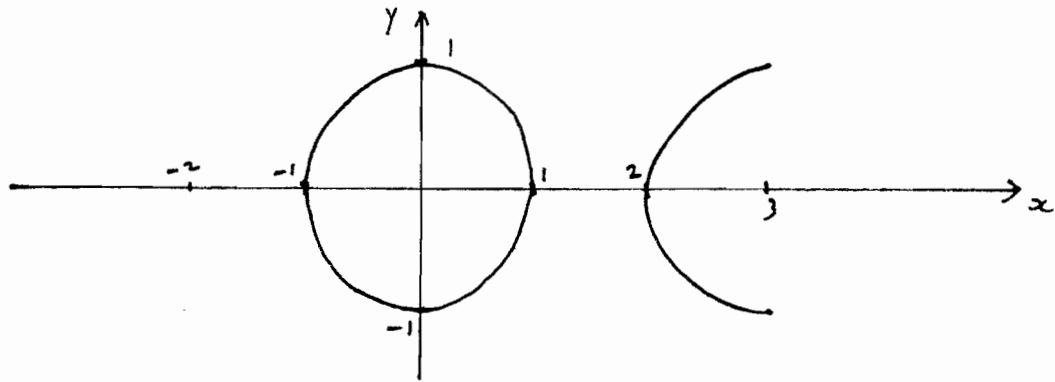
$$\underline{-4 < k < 4}$$

Q2. a. $\int \frac{x+1}{\sqrt{x-1}} dx$

$$= \int \frac{x-1}{\sqrt{x-1}} + \frac{2}{\sqrt{x-1}} dx$$
$$= \int (x-1)^{\frac{1}{2}} + 2(x-1)^{-\frac{1}{2}} dx$$
$$= \frac{2}{3} (x-1)^{\frac{3}{2}} + 4\sqrt{x-1} + c$$



iii)



c i) $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$

let $\cos \theta = x$

$\therefore 16x^4 - 16x^2 + 1 = 0$

$16 \cos^4 \theta - 16 \cos^2 \theta + 1 = 0$

$2(8 \cos^4 \theta - 8 \cos^2 \theta + 1) - 1 = 0$

$2 \cos 4\theta - 1 = 0$

$\cos 4\theta = \frac{1}{2}$

$4\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$

$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}$

$\therefore x = \cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, \cos \frac{7\pi}{12}, \cos \frac{11\pi}{12}$ (or equivalent)

ii) product of roots = $\frac{1}{16}$

$\therefore \cos \frac{\pi}{12} \cos \frac{5\pi}{12} \cos \frac{7\pi}{12} \cos \frac{11\pi}{12} = \frac{1}{16}$

but $\cos \frac{7\pi}{12} = -\cos \frac{5\pi}{12}$

$\cos \frac{11\pi}{12} = -\cos \frac{\pi}{12}$

$\therefore \cos^2 \frac{\pi}{12} \cos^2 \frac{5\pi}{12} = \frac{1}{16}$

$\cos \frac{\pi}{12} \cos \frac{5\pi}{12} = \pm \frac{1}{4}$ (but all angles in 1st quadrant)

$\therefore \cos \frac{\pi}{12} \cos \frac{5\pi}{12} = \frac{1}{4}$

iii)

$$16x^4 - 16x^2 + 1 = 0$$

$$\text{let } u = x^2$$

$$16u^2 - 16u + 1 = 0$$

$$\therefore u = \frac{16 \pm \sqrt{16^2 - 4 \times 16 \times 1}}{32}$$

$$= \frac{1}{2} \pm \frac{1}{4} \sqrt{3}$$

$$\therefore x^2 = \frac{1}{2} \pm \frac{1}{4} \sqrt{3}$$

$$\therefore x = \pm \sqrt{\frac{1}{2} \pm \frac{1}{4} \sqrt{3}}$$

$$\therefore \cos \frac{\pi}{12} = \sqrt{\frac{1}{2} + \frac{1}{4} \sqrt{3}} \quad (\text{largest all positive})$$

Q3.

a. $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$

$$= \int_0^1 \frac{\frac{2dt}{1+t^2}}{2 + \frac{1-t^2}{1+t^2}}$$

$$= \int_0^1 \frac{2dt}{t^2 + 3}$$

$$= \left[\frac{2}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$= \frac{\pi}{3\sqrt{3}}$$

$$t = \tan \frac{x}{2}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$dx = \frac{2dt}{1+t^2}$$

b. $P(x) = 4x^3 + 4x^2 - 7x + 2$

$$P'(x) = 12x^2 + 8x - 7$$

double root is a root of $P'(x) = 0$

$$12x^2 + 8x - 7 = 0$$

$$(6x+7)(2x-1) = 0$$

$$P\left(\frac{1}{x}\right) = 0$$

$\therefore x = \frac{1}{2}$ is double root

$\therefore$ roots are $\frac{1}{2}, \frac{1}{2}, \alpha$

$$\text{sum of roots} = -1$$

$$\frac{1}{2} + \frac{1}{2} + \alpha = -1$$

$$\alpha = -2$$

$\therefore$ Solutions $\frac{1}{2}, \frac{1}{2}, -2$

c. i) $P\left(\frac{x}{3}\right) = 0$

$$\left(\frac{x}{3}\right)^3 + P\left(\frac{x}{3}\right) + 9 = 0$$

$$\frac{x^3}{27} + \frac{px}{3} + 9 = 0$$

$$x^3 + 9px + 279 = 0$$

ii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$

$$= \frac{\delta(\alpha + \beta + \delta - \delta)}{\alpha\beta\delta}$$

$$\alpha + \beta + \delta = 0$$

$$\alpha\beta\delta = -9$$

$$= \frac{-\delta^2}{-9}$$

$$= \frac{\delta^2}{9}$$

iii) $\frac{1}{\alpha} + \frac{1}{\beta}, \frac{1}{\beta} + \frac{1}{\delta}, \frac{1}{\alpha} + \frac{1}{\delta}$

is equivalent to (from above)

$$\frac{\delta^2}{9}, \frac{\alpha^2}{9}, \frac{\beta^2}{9}$$

∴ required polynomial is

$$P(\sqrt{qx}) = 0$$

$$(\sqrt{qx})^3 + p\sqrt{qx} + q = 0$$

$$(\sqrt{qx})^3 + p\sqrt{qx} = -q$$

$$((\sqrt{qx})^3 + p\sqrt{qx})^2 = q^2$$

$$q^3 x^3 + 2pq^2 x^2 + p^2 qx - q^2 = 0$$

d. i)
$$\begin{aligned} \text{LHS} &= (1-x)^{n-1} - (1-x)(1-x)^{n-1} \\ &= (1-x)^{n-1} [1 - (1-x)] \\ &= x(1-x)^{n-1} \\ &= \text{RHS} \end{aligned}$$

ii)
$$\begin{aligned} I_n &= \int_0^1 \sqrt{x} (1-x)^n dx & u &= (1-x)^n & v &= \frac{2}{3} x^{\frac{3}{2}} \\ & & u' &= -n(1-x)^{n-1} & v' &= x^{\frac{1}{2}} \\ &= \cancel{\frac{2}{3} x^{\frac{3}{2}} (1-x)^n} \Big|_0^1 + \frac{2}{3} n \int_0^1 x^{\frac{1}{2}} (1-x)^{n-1} dx \\ &= \frac{2}{3} n \int_0^1 \sqrt{x} x (1-x)^{n-1} dx \\ &= -\frac{2}{3} n \int_0^1 \sqrt{x} (1-x) (1-x)^{n-1} dx + \frac{2}{3} n \int_0^1 \sqrt{x} (1-x)^{n-1} dx \\ & & & \text{(using part i)} \\ &= -\frac{2}{3} n I_n + \frac{2}{3} n I_{n-1} \end{aligned}$$

$$\therefore I_n = -\frac{2}{3}n I_n + \frac{2}{3}n I_{n-1},$$

$$\left(\frac{2}{3}n+1\right) I_n = \frac{2}{3}n I_{n-1}$$

$$\frac{2n+3}{3} I_n = \frac{2n}{3} I_{n-1}$$

$$I_n = \frac{2n}{2n+3} I_{n-1}$$

$$\text{iii) } I_3 = \int_0^1 \sqrt{x} (1-x)^3 dx$$

$$= \frac{6}{9} I_2$$

$$= \frac{6}{9} \left[\frac{4}{7} I_1 \right]$$

$$= \frac{6}{9} \cdot \frac{4}{7} \left[\frac{2}{5} I_0 \right]$$

$$= \frac{6}{9} \cdot \frac{4}{7} \cdot \frac{2}{5} \int_0^1 \sqrt{x} dx$$

$$= \frac{6}{9} \cdot \frac{4}{7} \cdot \frac{2}{5} \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^1$$

$$= \frac{6}{9} \cdot \frac{4}{7} \cdot \frac{2}{5} \cdot \frac{2}{3}$$

$$= \frac{32}{315}$$

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$